Example 1 - Milgram's Shocking Obedience Experiments

Social psychologist Stanley Milgram conducted a series of experiments to explore the nature of obedience

Milgram's premise was that people would often go to great and sometimes dangerous, or even immoral, lengths to obey an authority figure

Example 1 - Milgram's Shocking Obedience Experiments

Subjects were ordered to deliver increasingly strong electrical shocks to another person

While the person was an actor who was pretending, the subjects fully believed the other person was being shocked

Example 1 - Milgram's Shocking Obedience Experiments

The voltage levels started out at 30 volts and increased in 15-volt increments up to a maximum of 450 volts

The switches were also labeled with phrases including "slight shock," "medium shock," and "danger: severe shock."

The maximum shock level was simply labeled with an ominous "XXX."

Example 1 - Milgram's Shocking Obedience Experiments

Many participants were willing to deliver the maximum level of shock, even when the person pretending to be shocked was begging to be released or complaining of a heart condition

And while this result is very interesting the experiment itself caused considerable distress for the participants involved

Example 2 - Stanford Prison Experiment

- Aim: to see whether prison brutality is a result of the personality traits inherent in prisoners and guards, or is a consequence of the social environment in prisons

- Simulated prison environment where some participants acted as prisoners, some acted as guards

Example 2 - Stanford Prison Experiment

The “guards” began to torture the “prisoners”

- The “prisoners” began to show signs of psychological distress
- The “prisoners” wanted to leave and asked the “guards” for “parole” (they denied it)
- It ended on day 6 (of a planned 14) having descended into absolute chaos
- The scientists behind the experiment simply watched for the most part

Example 2 - Stanford Prison Experiment

“The experiment was the right thing to do, the wrong thing was to let it go past the second day”

- Was how one of the lead researchers later reflected on the work

Looking at history provides us evidence that we need to conduct research ethically

Laws and Regulations

Nowadays, ethics is embodied in laws and regulations:

- Computer Misuse Act, Health & Safety, GDPR, Intellectual Property, etc.

Computing scientists are expected to follow codes of ethics:

- ACM Code of Ethics: https://www.acm.org/code-of-ethics
- Etc.

Laws and Regulations

Researchers are also expected to adhere to codes of research ethics, especially when people are involved:

- BPS Code of Human Research Ethics:
  - https://www.bps.org.uk/news-and-policy/bps-code-human-research-e thics-2ndedition-2014

- EPSRC has various code of research ethics

- If involving animals, people with limited mental capacity, children, also specialist ethics and laws
  - Safeguarding is needed especially with vulnerable groups

Ethical Approval — Key Points From Glasgow's Requirements

- Participants must be able to give informed consent
- Participants have the right to withdraw at any time
- Participants have a right to know what will happen to their data
- Participants must be given contact details for the researchers

Ethical Approval at the University of Glasgow

At University of Glasgow, ethical approval is required for any experiment involving human participants or data:

- Submit an application describing the experiment, outlining potential ethical issues, and how you have addressed them
- Requires declaration you've adhered to the ESRC or BPS frameworks
- 1-3 members of the ethics committee review all proposals

Ethical Approval — Key Points From Glasgow's Requirements

- Participants should be informed if the work is commercially funded
- Data should maintain confidentiality
- The research must be in accordance with ESRC/BPS code of ethics

Ethics Throughout the Research Process

Ethical behaviour is necessary throughout the whole research process

- Not just when dealing with humans or their data

Committee on Publication Ethics (https://publicationethics.org/)

- Focuses on issues relating to research publication
- E.g., authorship, conflicts of interest, transparent peer review, data fabrication, etc.

Ethics Throughout the Research Process

Retraction Watch: https://retractionwatch.com/

- A blog highlighting retracted scientific publications
- Concerns over validity, ethical violations, falsification, misconduct, plagiarism, etc.
- More recently a lot of concerns about LLM causing plagiarism and false results/comments (especially images)
  - ACM now requires people to state if they have used LLM similar to academia

The importance of informed consent

Example - Facebook Mood Experiment

Based on a paper entitled: “Experimental evidence of massive-scale emotional contagion through social networks”

- The experiment hid certain elements from 689,003 peoples' news feed (about 0.04% of users) over the course of one week in 2012

  Found that it could make people feel more positive or negative through a process of "emotional contagion"

Paper link: Paper link: https://www.pnas.org/doi/10.1073/pnas.1320040111

Example - Facebook Mood Experiment

The important thing to note here is:

- Facebook did this without users' knowledge or explicit consent

- They effectively impacted the emotional state of 0.04% of its users for a week without their knowledge/consent

Informed consent is very important in an experiment

Consent is extremely important in the context of an experiment:

- As a rule of thumb, if your experiment involves human participants, their consent is required before they can be included in an experiment
- Should also be clear that they can revoke this consent and stop participating in the experiment at any time

Research into Ethics

Research Ethics in CS

- E.g. Matthew Chalmers here at Glasgow has done work in this space
- In this paper: Identified 2 main dimensions of participant ‘risk’:
  - Anonymous vs identifiable
  - User expectation of data access

Categorised Ethical Guidelines for Large Scale Mobile HCI

Donald McMillan
Mobile Life Centre @ SICS
SE-164, Kista, Sweden
don@mobilelifecentre.org

Alistair Morrison
School of Computing Science
University of Glasgow, UK
alistair.morrison@glasgow.ac.uk

Matthew Chalmers
School of Computing Science
University of Glasgow, UK
matthew.chalmers@glasgow.ac.uk

ABSTRACT
The recent rise in large scale trials of mobile software using ‘app stores’ has moved current researcher practice beyond available ethical guidelines. By surveying this recent and growing body of literature, as well as established professional principles adopted in psychology, we propose a set of ethical guidelines for large scale HCI user trials. These guidelines come in two parts: a set of general principles and a framework into which individual app store-based trials can be assessed and ethical concerns exposed. We categorise existing literature using our scheme, and explain how researchers could use our framework to classify their future user trials to determine ethical responsibility, and the steps required to meet these obligations.

Author Keywords
Ethics; large-scale trials; mass participation; app stores.

ACM Classification Keywords
H.5.2. User Interfaces: Evaluation/methodology

INTRODUCTION
Large scale trials of mobile HCI systems using ‘app stores’ are becoming increasingly popular [9], yet it has been noted that such trials raise non-trivial ethical challenges [5]. Issues such as competing sets of guidelines, differences in international laws and research cultures, and lack of community consensus can leave researchers unsure as to how to run a study which meets their ethical obligations.

Perhaps the best-known guidelines specific to mobile and ubiquitous computing are those in Greenfield’s Everyware book [18]. High-level guidelines such as ‘do no harm’ and ‘default to harmlessness’ were discussed, and are still generally applicable, but have yet to be contextualised to suit new ubicomp research practices. Emerging user practices, such as the widespread use of web sites such as YouTube and Facebook, and the near-ubiquity of cameras on phones also make more established guidelines, e.g. in MacKay’s CHI ’95 Ethics, Lies and Videotape paper [23] seem rather outdated. People are increasingly accustomed to the dissolution of traditional social barriers of privacy, driven by the poor privacy controls commonly provided by online social networking sites [36]. However, this shift in user attitude cannot be expected to be consistent across, or even within, cultures and demographics. The possible harm to the reputation of the researcher, or research as a whole, by overestimating this movement in opinion outweighs the benefits of taking a too relaxed attitude to the ethical diligence required by researchers.

In this paper, we look at existing sets of ethical principles for human trials, making a case for how they could be interpreted to encompass large scale mobile software trials. We then identify two key ethical issues for mobile HCI: anonymisation of participants and user understanding/expectation of data logging. We create a four-category classification of ethical requirements based on these factors. This is illustrated by surveying recent literature from this field and assigning past studies to one of our created categories. Finally, we take a two-stage approach to proposing a set of ethical guidelines for mobile HCI. Firstly, we provide a set of general principles based on our interpretations of the existing guidelines and secondly, we provide guidelines for each of the quadrants identified in our framework.

EXISTING ETHICAL GUIDELINES
Increasingly, HCI research crosses institutional, professional and national boundaries, further complicating the application of appropriate ethics protocols and review processes. For these reasons, researchers’ development of detailed and specific regulations on the handling of ethics issues in HCI research, with the aim of covering all eventualities, is seen by many ethicists as an ultimately flawed direction [14]. As soon as one new set of regulations is finalised, a new method or topic of research is likely to emerge that is not covered. The existence of lengthy, detailed and prescriptive professional or institutional regulations raises the risk of researchers following the letter, but not the spirit, of the regulations and may in consequence lead to research being carried out that is ethically flawed.

HCI is by no means the only field of research which uses human trials as a method of evaluating hypotheses and exploring ideas. Notably, the fields of Psychology and Medicine have well established guidelines compiled and upheld by professional bodies. While it may be argued that the potential harm of an ill-run medical or psychological trial is much greater than that posed by a piece of mobile or online research, significant concerns are raised by the increasing value of personal data, the volume of such data it is possible to collect and the difficulties surrounding anonymisation.

Finally, a note about doing research

Skills needed to do research

As a scientist / researcher you must wear many hats

- You are required to do a number of functionally different skills/tasks as part of the research process
- Conducting a literature review is a different skill than designing an experiment which is a different skill than analysing the data, etc.

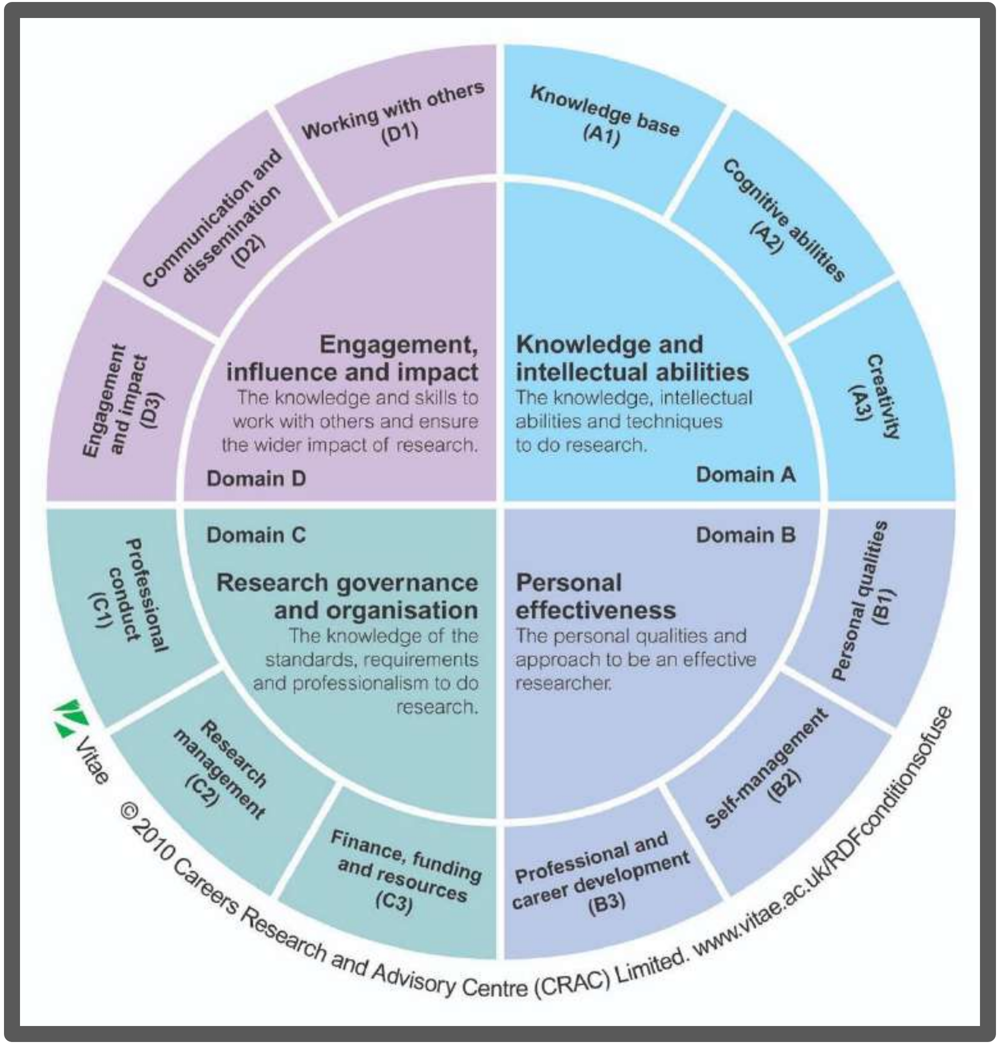
You can specalise in a single part of the process, but often you will do it all.

A summary of the different stages of the research process

1	Read and understand a paper.	Skim, then read detail	Understanding
2	Summarise the content of a paper.	Demonstrate that you know what the paper is about: its key points and conclusions	Annotated bibliography
3	Explain how different papers relate to each other	Agreements, contradictions, extensions, replications	Literature review
4	Identify a gap in the literature	Useful, relevant, important	Idea
5	State a RQ arising from this gap	Clear question (with a ? at the end)	RQ
6	Devise and justify a research plan to address the RQ	Feasible, systematic	Project proposal
7	Get the resources	Appropriate funding bodies	Funding application
8	Execute the research plan	'Do' the research, collaborate, supervise, attend conferences	Data, frameworks, models, systems, protocols, algorithms...
9	Write and publish papers	Journals, conferences	Papers, presentations
10	Critically review a paper	Peer-review; discussion	Critique, suggestions, acceptance decision

The Vitae framework

The Vitae Researcher Development Framework was designed for planning, promoting and supporting the personal, professional and career development of researchers in higher education.



Any Questions?

Summary of Lecture 3 Part 2: Lecture

- We spoke about hypothesis
  - What it is, how to form one and examples
- Ethics
  - Why we need it, and examples of poor ethical behaviour
  - Importance of ethics within computer science

Next Week: Experimental Design, two videos to watch before lecture